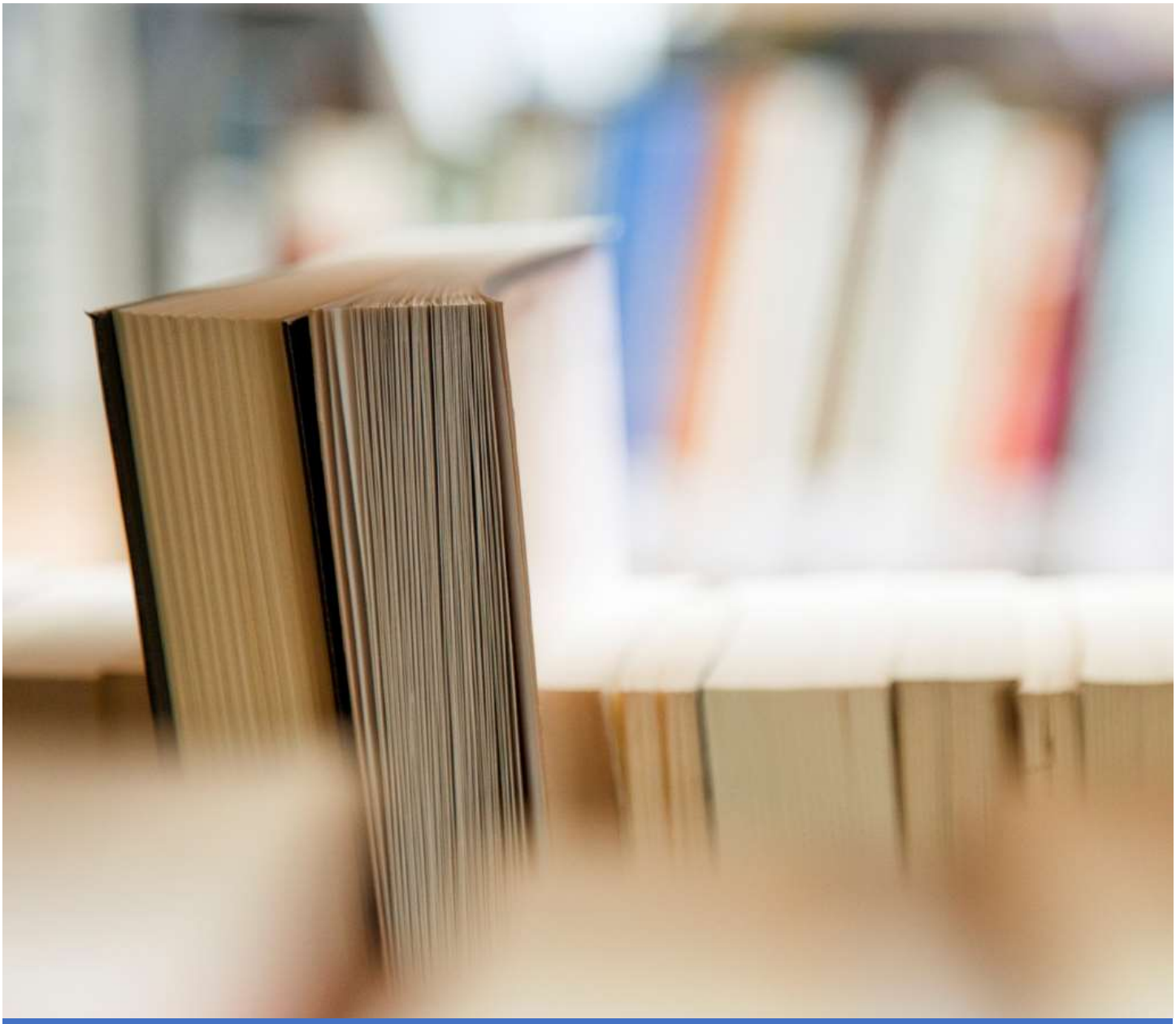


AGILE

LAB ASSESSMENT

23MIS0091 B. SANJAY AGILE FAT

1.CODE:

pom.xml | Agilefat.java 3 | Agilefattest.java 1

src > main > java > com > example > Agilefat.java > Agilefat > main(String[] args)

```java
package com.example;

public class Agilefat {
    static public String str1 = "M. adhavan-by adihesh";
    static public String str2 = "life-by ranveer";
    static public String str3 = "man-by gauresh";

    Run main | Debug main
    public static void main(String[] args) {

        String result_1 = listingLogic();
        System.out.println("Listing available books: \n");
        System.out.println(result_1);

        String result_2 = lending("M. adhavan-by adihesh");
        System.out.println("Lending: \n");
        System.out.println(result_2);

    }

    static public String listingLogic()
    {
        String result = str1+str2+str3;
        return result;
    }

    static public String lending(String book)
    {
        if (book == str1)
        {
            return (book+"-lended");
        }
        else if (book == str2)
        {
            return(book+"-lended");
        }
        else if (book == str3)
        {
            return(book+"-lended");
        }
        else
        {
```

1.2 CODE(TEST):

```
pom.xml    Agilefat.java 3    Agilefattest.java 3 ×
src > test > java > com > example > Agilefattest.java > Agilefattest > testlending3()
1  package com.example;
2
3  import static org.junit.jupiter.api.Assertions.assertEquals;
4  import org.junit.jupiter.api.Test;
5
6  public class Agilefattest {
7
8      Agilefat calculator = new Agilefat();
9
10     @Test
11     void testlending1()
12     {
13         assertEquals("man-by gauresh-lended", calculator.lending("man-by gauresh"));
14     }
15     @Test
16     void testlending2()
17     {
18         assertEquals("life-by ranveer-lended", calculator.lending("life-by ranveer"));
19     }
20     @Test
21     void testlending3()
22     {
23         assertEquals("M. adhavan-by adihesh-lended", calculator.lending("M. adhavan-by adihesh"));
24     }
25 }
26
```

1.3 RESULT:

```
PROBLEMS 10    OUTPUT    DEBUG CONSOLE    TERMINAL
PS D:\Users\balac\Spring Boot\agilefat> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:maven-calculator >-------------------
[INFO] Building maven-calculator 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ maven-calculator ---
[INFO] Deleting D:\Users\balac\Spring Boot\agilefat\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ maven-calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Users\balac\Spring Boot\agilefat\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ maven-calculator ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ maven-calculator ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
```

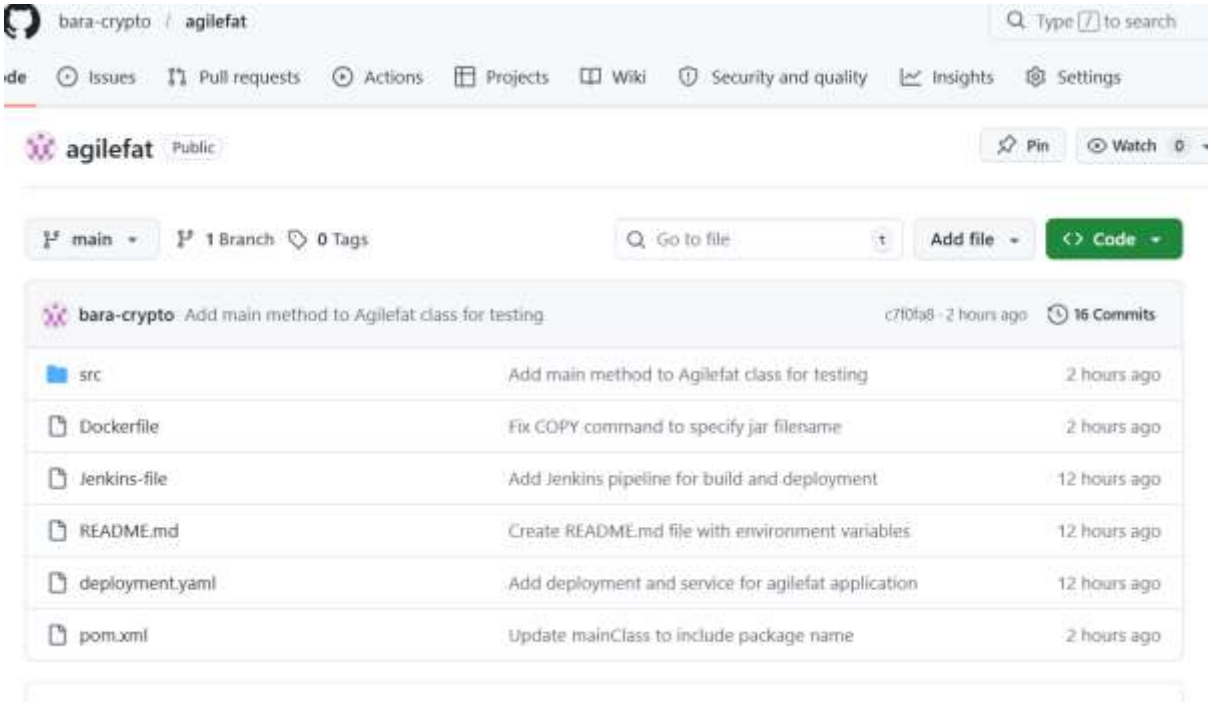
```
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ maven-calculator ---
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ maven-calculator ---
[INFO] Building jar: D:\Users\balac\Spring Boot\agilefat\target\agilefat.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.729 s
[INFO] Finished at: 2026-04-13T10:33:08+05:30
[INFO] ------------------------------------------------------------------------
```

2.1 GIT

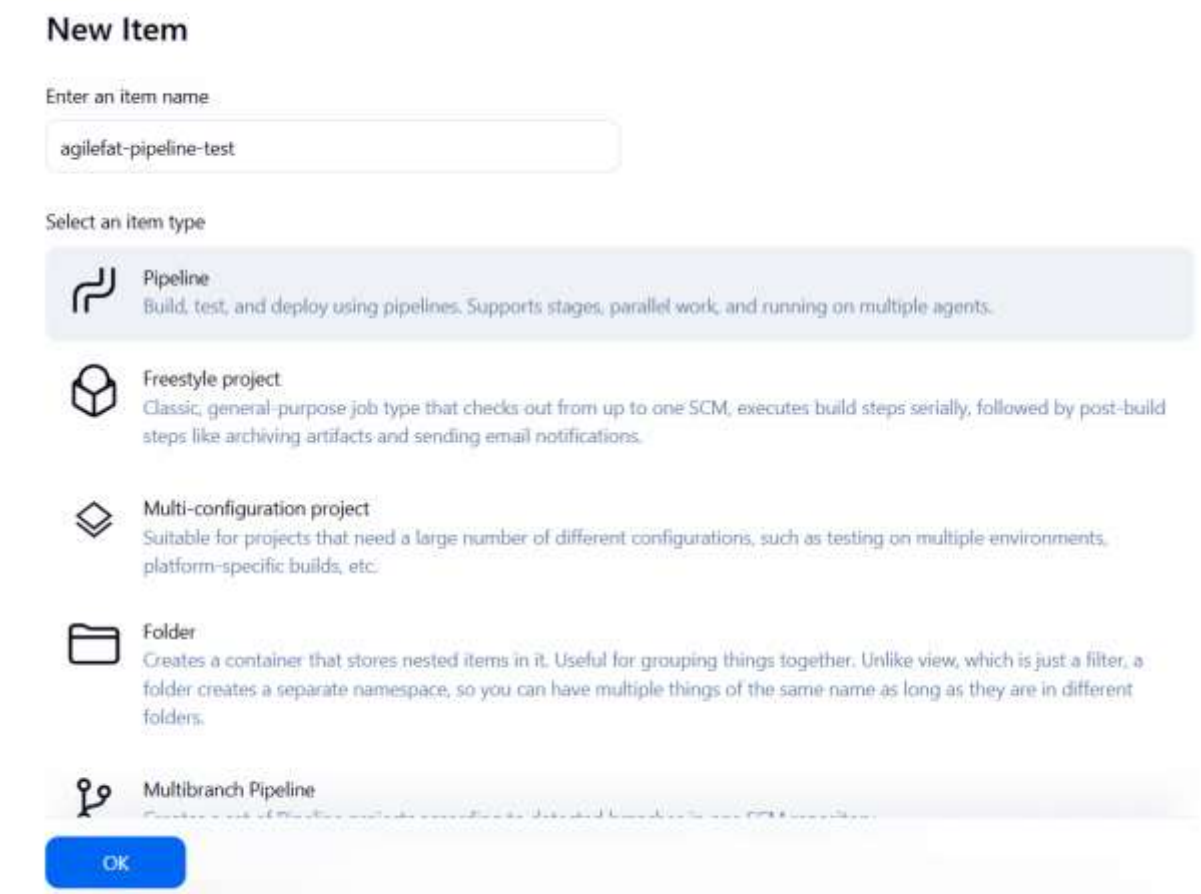
```
balac@Lenova MINGW64 /d/Users/balac/Spring Boot/agilefat (main)
$ git init
Reinitialized existing Git repository in D:/Users/balac/Spring Boot/agilefat/.gi
t/

balac@Lenova MINGW64 /d/Users/balac/Spring Boot/agilefat (main)
$ git add .
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the ne
xt time Git touches it
warning: in the working copy of 'Jenkins-file', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'deployment.yaml', LF will be replaced by CRLF t
he next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'src/main/java/com/example/Agilefat.java', LF wi
ll be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test/java/com/example/Agilefattest.java', L
F will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compi
le/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testC
ompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the ne
xt time Git touches it
```

```
balac@Lenova MINGW64 /d/Users/balac/Spring Boot/agilefat (main)
$ git branch -M main

balac@Lenova MINGW64 /d/Users/balac/Spring Boot/agilefat (main)
$ git add origin main https://github.com/bara-crypto/agilefat
```

2.2 GITHUB:



3. JENKINS:



3.1. JENKINS PIPELINE:

```
13    stages {
14        stage('Initialize & Clone') {
15            steps {
16                git branch: 'main', url: "${REPO_URL}"
17            }
18        }
19
20        stage('Maven Build') {
21            steps {
22                // This produces target/agilefat.jar
23                bat 'mvn clean package -DskipTests'
24            }
25        }
26
27        stage('Docker Image Build & Push') {
28            steps {
29                withCredentials([usernamePassword(credentialsId: 'docker-creds', passwordVariable: 'DOCKER_PASS', us
30                    bat "docker login -u %DOCKER_USER% -p %DOCKER_PASS%"
31                    bat "docker build -t sanjaydocker2005/agilefat:latest ."
32                    bat "docker push sanjaydocker2005/agilefat:latest"
33                }
34            }
35        }
36
37        stage('Kubernetes Deployment') {
```

```
36
37        stage('Kubernetes Deployment') {
38            steps {
39                bat 'kubectl apply -f deployment.yaml'
40            }
41        }
42    }
43 }
```

Use Groovy Sandbox ?

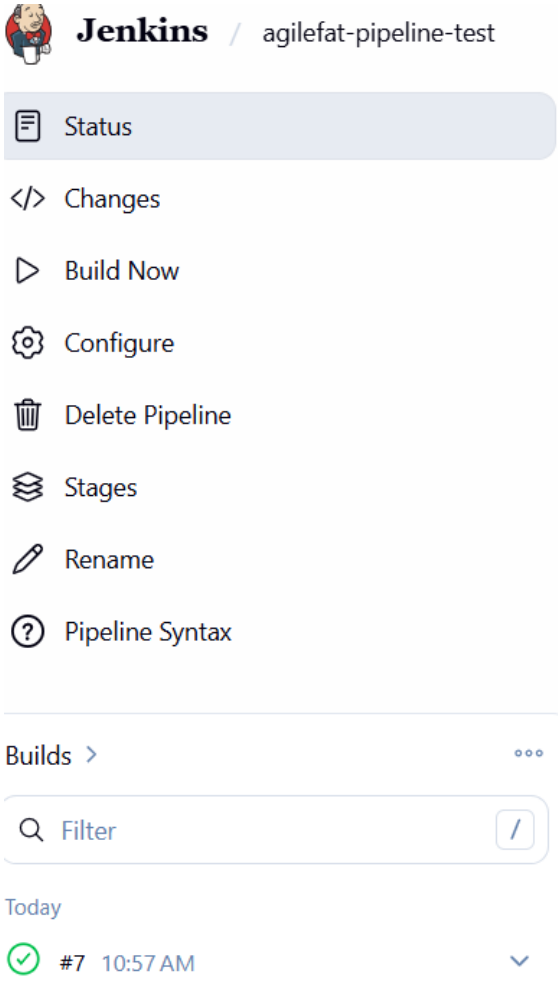
ipeline Syntax

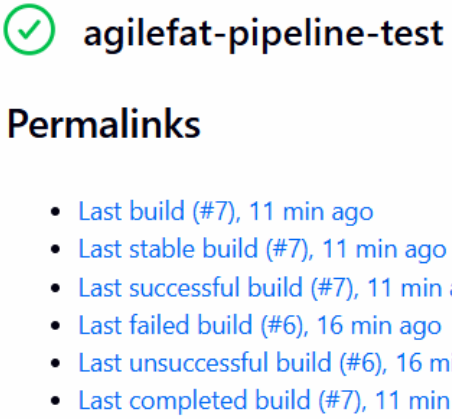
nced

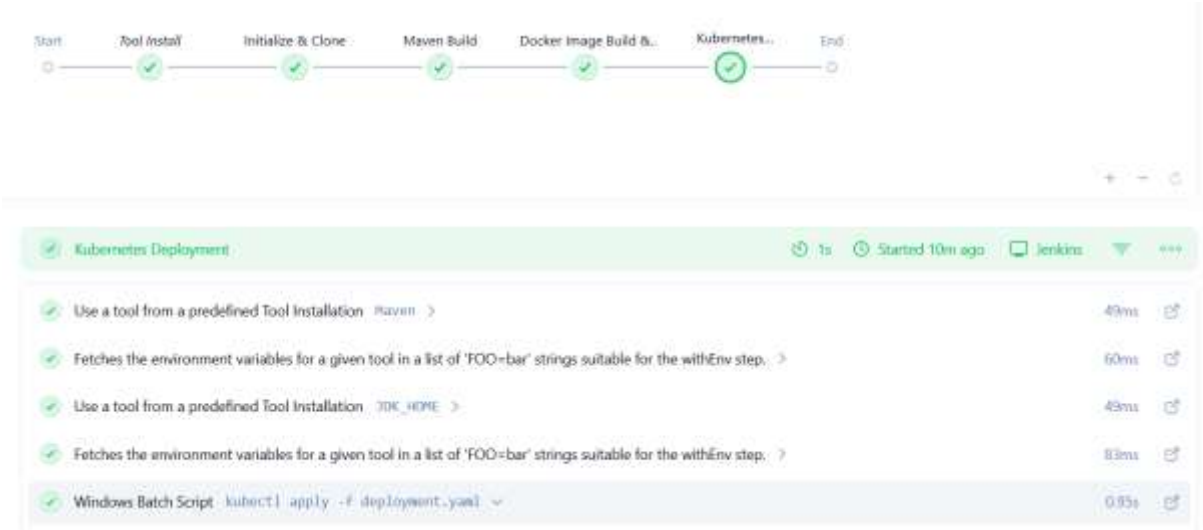
anced

Save Apply

4. OUTPUT:


Jenkins / agilefat-pipeline-test
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
Builds
Filter
/
Today
#7 10:57 AM


agilefat-pipeline-test
Permalinks
Last build (#7), 11 min ago
Last stable build (#7), 11 min ago
Last successful build (#7), 11 min ago
Last failed build (#6), 16 min ago
Last unsuccessful build (#6), 16 min ago
Last completed build (#7), 11 min ago


Start
Tool Install
Initialize & Clone
Maven Build
Docker Image Build &...
Kubernetes...
End
Kubernetes Deployment
1s
Started 10m ago
Jenkins
Use a tool from a predefined Tool Installation Maven
49ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
60ms
Use a tool from a predefined Tool Installation JDK_HOME
49ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
83ms
Windows Batch Script kubectl apply -f deployment.yaml
0.95s

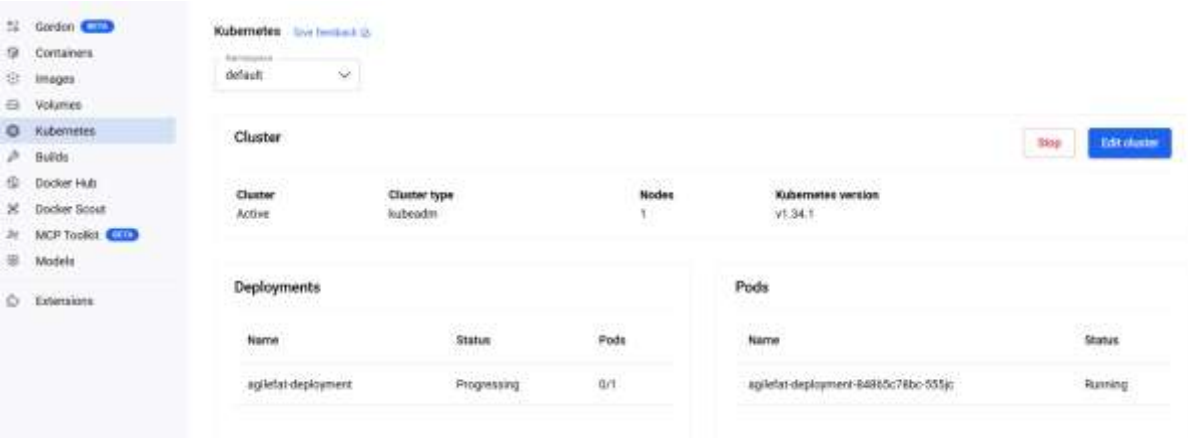
Console

```
Started by user sanjay
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\balac\.jenkins\workspace\agilefat-pipeline-test
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Initialize & Clone)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
```

```
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\cmd\git.exe rev-parse --resolve-git-dir C:\Users\balac\.jenkins\workspace\a
Fetching changes from the remote Git repository
 > C:\Program Files\Git\cmd\git.exe config remote.origin.url https://github.com/bara-crypto/agilefa
Fetching upstream changes from https://github.com/bara-crypto/agilefat
 > C:\Program Files\Git\cmd\git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > C:\Program Files\Git\cmd\git.exe fetch --tags --force --progress -- https://github.com/bara-cryp
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > C:\Program Files\Git\cmd\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision a14858f50e6500abe3397dace5eed98f50a4cedc (refs/remotes/origin/main)
 > C:\Program Files\Git\cmd\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -f a14858f50e6500abe3397dace5eed98f50a4cedc # timeout=
 > C:\Program Files\Git\cmd\git.exe branch -a -v --no-abbrev # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -D main # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -b main a14858f50e6500abe3397dace5eed98f50a4cedc # tim
Commit message: "nw"
 > C:\Program Files\Git\cmd\git.exe rev-list --no-walk b874146c5591ccc2fa8230d158c214049ada3720 # t
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Maven Build)
[Pipeline] tool
[Pipeline] envVarsForTool
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.106 s
[INFO] Finished at: 2026-04-13T10:58:05+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Docker Image Build & Push)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withCredentials
Masking supported pattern matches of %DOCKER_PASS%
[Pipeline] {
[Pipeline] bat

C:\Users\balac\.jenkins\workspace\agilefat-pipeline-test>docker login -u sanjaydocker2005 -p ****
```

```
C:\Users\balac\.jenkins\workspace\agilefat-pipeline-test>docker build -t sanjaydocker2005/agilefat:latest .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 759B 0.0s done
#1 DONE 0.1s

#2 [auth] library/maven:pull token for registry-1.docker.io
#2 DONE 0.0s

#3 [auth] library/eclipse-temurin:pull token for registry-1.docker.io
#3 DONE 0.0s

#4 [internal] load metadata for docker.io/library/maven:3.8.5-openjdk-17
#4 ...

#5 [internal] load metadata for docker.io/library/eclipse-temurin:17-jre-alpine
#5 DONE 2.3s

#4 [internal] load metadata for docker.io/library/maven:3.8.5-openjdk-17
#4 DONE 2.4s

#6 [internal] load .dockerignore
#6 transferring context: 2B 0.0s done
#6 DONE 0.0s

#7 [build 1/5] FROM docker.io/library/maven:3.8.5-openjdk-17@sha256:3a9c30b3af6278a8ae0007d3a3bf00fff80ec3ed7ae4eb9bfa1772853101549b
#16 exporting attestation manifest sha256:aa9e4ae8d54f849774484dd41e21b72ad3a0d0ee0180bbe2b3ef1d3c8eeeb3 0.0s done
#16 exporting manifest list sha256:e2654b8389df56e774c1b46d5881e944f59dd06fbc3c0154d95cf2664d2173aa
#16 exporting manifest list sha256:e2654b8389df56e774c1b46d5881e944f59dd06fbc3c0154d95cf2664d2173aa 0.1s done
#16 naming to docker.io/sanjaydocker2005/agilefat:latest 0.0s done
#16 unpacking to docker.io/sanjaydocker2005/agilefat:latest 0.1s done
#16 DONE 0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/e3tbexg738pf34p0absil4ynm
[Pipeline] bat

C:\Users\balac\.jenkins\workspace\agilefat-pipeline-test>docker push sanjaydocker2005/agilefat:latest
The push refers to repository [docker.io/sanjaydocker2005/agilefat]
b50a56d2ab38: Waiting
589002ba0eae: Waiting
8a88a697ab44: Waiting
2b28c8d76488: Waiting
4adb00321f3d: Waiting
3958e9165451: Waiting
27bf0477d85a: Waiting
7cb49e20f90d: Waiting
7cb49e20f90d: Waiting
b50a56d2ab38: Waiting
```

```
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
27bf0477d85a: Waiting
7cb49e20f90d: Pushed
27bf0477d85a: Pushed
latest: digest: sha256:e2654b8389df56e774c1b46d5881e944f59dd06fbc3c0154d95cf2664d2173aa size: 856
latest: digest: sha256:e2654b8389df56e774c1b46d5881e944f59dd06fbc3c0154d95cf2664d2173aa size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Kubernetes Deployment)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] bat

C:\Users\balac\.jenkins\workspace\agilefat-pipeline-test>kubectl apply -f deployment.yaml
deployment.apps/agilefat-deployment unchanged
service/agilefat-service unchanged
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

4. DOCKER & KUBERNETES:

4.1 KUBERNETES: